

Neel Ahuja

(973) 382-3534 | ahuja@stanford.edu | [LinkedIn](#) | [Google Scholar](#) | [Github](#)

EDUCATION

Stanford University, B.S. in Computer Science, Artificial Intelligence Track *Expected Graduation June 2028*

Relevant Coursework: Programming Abstractions, Computer Systems, Linear Algebra & Multivariable Calculus, Probability for CS

Achievements: Regeneron STS Top 300, ISEF 2nd Place Grand Award, National JSHS 1st Place, Gold Presidential Service Award

Organizations: Stanford AI Lab (SAIL), Stanford Consulting, Conservation Technology, Stanford Climate Week, Dorm President

WORK EXPERIENCES

SALT Lab @ Stanford Artificial Intelligence Lab (SAIL) *Palo Alto, CA*

AI Research Intern | Python, PyTorch, MongoDB, Docker, MkDocs *Dec 2025 – Present*

- **Built automated NLP benchmarking framework evaluating 4 AI agents** (Codex, Gemini-CLI, Claude Code, Manus) across 350+ real-world workflow tasks, using rubric-based LLM judging and structured evaluation metrics
- Designed ranking system with MongoDB backend and RAG automated grading, **enabling standard model evaluation of new AI agents** by analyzing delivery rates, reasoning steps, and output patterns, enhancing human-AI collaboration
- **Developed and maintained technical documentation using MkDocs for CoCo (Collaborative Cognitive Agent)**, a 40,000+ line monorepo enabling human-agent workflow collaboration, reducing developer onboarding time by 60%

New Jersey Institute of Technology Zhang Laboratory

Newark, NJ

Research Intern | MATLAB, R, Prism, Excel

June 2023 – June 2025

- **Co-authored 2 published peer-reviewed studies** by designing and implementing a novel water purification solution using membranes and hydroponics plants, removing 71.1% of contaminants from a local pond at 95% lower cost than alternatives
- **Analyzed 250 experiment trials with statistical tests** (ANOVA, regression) in MATLAB, processing 1000+ data points into publication-ready figures using Prism/Excel, and **presented results at 10 research conferences**
- **Pitched solution to EPA officials** by demonstrating \$10M+ in potential government water purification cost savings

Newtristics

Remote

Machine Learning Intern | Python, Numpy, PyTorch, Figma

June 2021 - Nov 2022

- **Built a random forest supervised learning model with 90% accuracy to classify 56 heuristics** (psychological biases) in marketing messages and assist copywriters, led 12 writers & leveraged GenAI to produce 5000+ synthetic training messages
- **Scored 1000+ generated marketing messages from 6 leading LLM copywriting tools** using NLP sentiment analysis algorithms, then proposed a multimodal integration strategy to improve generated message **quality, speed, and scalability for 50+ clients**

PROJECTS

FALCON: Fraud & Anomaly Learning Classifier for On-chain Networks | Numpy, PyTorch, Pandas *June 2025 – Sept 2025*

- **Built XGBoost deep learning classifier model** to identify fraudulent Ethereum transactions with 92% accuracy, validated performance with confusion matrix & stratified train-test splits, reducing loss by tuning hyperparameters (learning rate, etc.)
- **Minimized overfitting** using cross-validation, L2 regularization & early stopping by generating training graphs with Matplotlib
- Deployed fraud detection model on a **website with 20+ users**, enabling real-time classification of 100+ blockchain transactions

Millburn School Security Kiosks | Java, JavaScript, React, WebKit, Django, Android Studio

Sept 2021 - Nov 2022

- Deployed 8 student sign-in kiosks with **200+ daily users** by **building a frontend Android app** for student ID authentication and a **Django + SQL backend** with real-time logs for security guard to monitor students entering/exiting
- Enhanced school security by **integrating FaceNet facial recognition** (95% accuracy) into kiosks using NVIDIA Jetson Nano, trained a recognition model on 1,200 student ID photos and **secured it with AES encryption**
- **Secured \$2,000 in district funding** for facial recognition solution by pitching privacy-focused guardrails to Board of Education

ML-Optimized Enzyme With Plastic Degrading Properties | National JSHS 1st Place | Pandas, Flask

Oct 2022 – May 2023

- Designed a plastic-degrading enzyme by building an **unsupervised learning ensemble model of CNN and RNN** from literature
- Simulated 6,517 mutations, determining 5 optimal genetic edits using **MCDA statistical test** to improve protein stability by 10%
- **Published in a peer-reviewed journal** and **won 1st place** at the Department of Defense JSHS competition

Jersey Cares Social Media Mobile App | Gold Presidential Service Award | Dart, Firebase, JSON

Sept 2021 - Sept 2022

- Built a volunteer **social media outreach app** intended for **300+ Jersey Cares volunteers** in **Flutter + Firebase**, implementing Google OAuth sign-in, and integrating Firestore for dynamic posts, messaging, and event storage.
- Automated event discovery for 50+ volunteer opportunities through web scraping and **push notification system using Firebase**

TECHNICAL SKILLS

Programming Languages: Python, C++, C, Java, Javascript, R

Frameworks & Tools: PyTorch, Numpy, Tensorflow, Pandas, React, MongoDB, Docker, CAD, Firebase, JSON, Django, SQL